HASAN AHMED

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EDUCATION

Ph.D. in Geography and Environmental Systems – Spatial Ecology Focus 2021 – Present
University of Maryland, Baltimore County – Baltimore, MD

M.S. in Geo-information Science and Earth Observation (Natural Resources) 2021
University of Twente. (Faculty ITC) – Enschede, The Netherlands

M.S. in Forestry (with distinction) 2018
Shahjalal University of Science and Technology, Sylhet, Bangladesh

B.S. (Honours) in Forestry 2016
Shahjalal University of Science and Technology, Sylhet, Bangladesh

EXPERIENCE

Graduate Assistant (Teaching & Research*)* Aug 2021 – Present
University of Maryland, Baltimore County (UMBC) – Baltimore, MD

- Reviewed, cleaned, and validated complex geospatial datasets; conducted accuracy assessments and produced documentation, metadata, and publication-ready maps for peer-reviewed research and stakeholder reports.
- Developed and implemented Python and Google Earth Engine scripts for land-cover mapping, forest disturbance detection, and spatial data processing.
- Contributed to NASA JPL and NASA LCLUC projects involving multi-source spatial data integration, data quality review, and the preparation of analytical summaries for research and applied use.
- Instructed and supported students in GIS, remote sensing, spatial statistics, environmental science, and field techniques; led lab sections and provided hands-on technical troubleshooting support.
- Provided technical guidance on geospatial data processing, spatial modeling, classification, accuracy assessment, and field-based data validation for student projects and coursework.

Graduate Research Fellow, Internship Nov 2024 – Feb 2025
NASA Jet Propulsion Laboratory (JPL) – Sub-orbital Radar Science and Engineering

- Contributed with GIS and Earth Observation analysis to a SERVIR Amazonia project to create land-cover maps using SAR and Optical Observations.
- Conducted advanced geospatial analysis to map agroforestry, perennial crops, stable forests, deforestation, and forest degradation.
- Reproduced and updated forest disturbance maps as input layers for the mapping workflows.
- Assessed the accuracy of land-cover and forest disturbance maps.
- Created Python scripts to support data preparation and integrate them into the crop mapping module.
- Collaborated with interdisciplinary scientists to translate technical results into actionable geospatial products for regional stakeholders.

Instructor Jan 2019 – Jul 2019
Excel Institute (Formerly FM Method)– Sylhet, Bangladesh

- Conducted IELTS preparation classes for students across varied proficiency levels in Listening, Reading, Writing, and Speaking.
- Prepared structured lesson plans, classroom activities, and practice materials aligned with IELTS exam requirements.

- Evaluated student progress through mock tests, writing assessments, and speaking practice, offering constructive feedback for improvement.
- Guided students in exam techniques, time management, and confidence building to support achievement of target band scores.

Research Assistant Jan 2018 – Dec 2018
University of Lisbon (Ulisboa), Lisbon, Portugal – Centre for Ecology, Environment, and Evolution

- Applied Earth observation analytics to detect and monitor mangrove and coastal land-cover change, supporting data-driven coastal management for climate change mitigation and conservation planning.
- Processed and analyzed multi-temporal satellite imagery to quantify ecosystem change and identify spatial patterns of degradation and recovery.
- Supported ecological research by integrating remote sensing outputs with field and ancillary environmental data.
- Produced maps, spatial summaries, and technical documentation to inform conservation planning and climate change mitigation strategies.

Instructor Jan 2017 – Jan 2018
Forestry Science and Technology Institute – Sylhet, Bangladesh

- Trained government forestry officers in GIS-based forest mapping, stand-level assessment, spatial planning, and integration of field inventory data with geospatial tools.
- Designed and supervised field-based inventory exercises with emphasis on measurement protocols, sampling design, data quality control, and interpretation for management and reporting.
- Developed instructional materials and technical documentation aligned with national forestry monitoring standards.

GIS Intern Oct 2015 – Dec 2015
Creative Conservation Alliance (CCA) – Dhaka, Bangladesh

- Developed species distribution maps for pythons across forested regions of Bangladesh, supporting wildlife conservation and habitat management efforts.
- Compiled and processed species occurrence records and environmental data to produce geospatial distribution outputs
- Applied spatial analysis and cartographic techniques to produce conservation-focused geospatial outputs.

PUBLICATIONS AND PRESENTATIONS

1. **Ahmed, H.** and M.E. Fagan. "Assessing Forest Structure and Biophysical Controls on Degradation Across the Amazon Using Spaceborne LiDAR." In ForestSAT 2026, Gainesville, FL, USA. 2026.

2. Fagan, M. E., Schwartz, N. B., Alonzo, M., **Ahmed, H.**, F. Saad. "The (mis)measurement of forest edge effects using remote sensing." In ForestSAT 2026, Gainesville, FL, USA. 2026.

3. **Ahmed, H.**, Fagan, M.E., and N.S. Pinto. "Comparing the performance of satellite-based forest degradation products in the Peruvian and Brazilian Amazon." In AGU Meeting 2024, Global Environmental Change, Washington DC, USA. 2024.

4. Saha, S, Uddin, M.B., and **H. Ahmed**. "Detection of Forest Cover Change and Spatial Prediction of Growing Stock in Satchari National Park, Bangladesh." In Young Researchers Meeting on Conservation and Sustainable Use of Forest Systems, 2023, Palencia, Spain. Feb 2023.

5. **Ahmed, H.**, Pinto, N.S., and M.E. Fagan. "Evaluating the Sensitivity of Spectral and Synthetic Aperture Radar-based Forest Degradation Products in the Peruvian Amazon Forest." In AGU Fall Meeting 2022, Biogeosciences, Chicago, IL, USA. 2022. DOI: 10.1002/essoar.10512892.31.

6. **Ahmed, H.** "A comparison between UAV-RGB and ALOS-2 PALSAR-2 images for the assessment of aboveground biomass in a temperate forest." Master's thesis, University of Twente, Enschede, Netherlands. 2021.

7. Rahman, M.R., Sajib, E.H., Chowdhury, I.M., Banik, A., Bhattacharya, R., and **H. Ahmed**. "Present scenario of COVID-19 in Bangladesh and government preparedness for facing challenges." J Adv Biotechnol Exp Ther 4, no. 2 (2021): 187-199.

8. Roy, B., **Ahmed, H.**, and G.P. Penha-Lopes. "Morphological change detection caused by seasonal monsoon and cyclones in the coast over time: a case study on Bangladesh delta." In EGU General Assembly Conference Abstracts, p. 1469. 2019.

9. **Ahmed, H**. "Assessing spatio-temporal changes in forest structural attributes at Lawachara National Park, Bangladesh." Master's thesis, Shahjalal University of Science and Technology, Sylhet, Bangladesh. 2018.

10. **Ahmed, H.** "Detecting tree richness hotspots using Species Area Relationships to compare with land-covers and forest composition of Rema-Kalenga Wildlife Sanctuary, Habiganj, Bangladesh." Bachelor's thesis, Shahjalal University of Science and Technology, Sylhet, Bangladesh. 2016.

WORKSHOPS & TRAININGS

1. **Spaceborne LiDAR for Monitoring Vegetation Structure and Biomass using GEDI.** NASA Applied Remote Sensing Training Program. November 2025

2. **An Introduction to Synthetic Aperture Radar (SAR) and Its Applications.** NASA Applied Remote Sensing Training Program. November 2024.

3. **Large Scale Applications of Machine Learning using Remote Sensing for Building Agriculture Solutions.** NASA Applied Remote Sensing Training Program. March 2024.

4. **Overview of SeaDAS 8.4.1 for the Processing, Analysis, and Visualization of Optical Remote Sensing Data for Water Quality Monitoring.** NASA Applied Remote Sensing Training Program. February 2024.

5. **Spectral Indices for Land and Aquatic Applications.** NASA Applied Remote Sensing Training Program. November 2023.

6. **Selecting Climate Change Projection Sets for Mitigation, Adaptation, and Risk Management Applications.** NASA Applied Remote Sensing Training Program. September 2022.

7. **Species Distribution Modeling with Remote Sensing.** NASA Applied Remote Sensing Training Program. August 2021.

8. **Neural Networks and Deep Learning.** DeepLearning.AI (through Coursera Online). February 2021.

9. **Tailor-made Training on the Use of UAV Images for the Application of Remote Sensing on Natural Resources.** Faculty ITC, University of Twente, Enschede, the Netherlands. April 2018.

10. **Workshop on Ecology, status, and conservation of Python and tortoise. Radio Tracking of Pythons and Tortoise.** Bangladesh Python Project. Lawachara National Park, Moulvibazar, Bangladesh. June 2014.

11. **Documentary Film Making and Nature Documentation.** Green Explore Society, Shahjalal University of Science and Technology, Sylhet, Bangladesh. March 2014.

12. **Bat Conservation and Research Techniques.** Chittagong Veterinary and Animal Science University, Chittagong, Bangladesh. January 2014

13. **Training on SPSS and MATLAB.** Department of Forestry and Environmental Science, Shahjalal University of Science and Technology, Sylhet, Bangladesh. April 2013.

14. **Workshop on Birds of Tanguar Haor: Species Diversity, Habitat Utilization and Conservation Initiatives through Local Community.** Shahjalal University of Science and Technology, Sylhet, Bangladesh. December 2012.

AWARDS & SCHOLARSHIPS

CAHSS Dean's Summer Fellowship ($ 6,000) 2025
For Doctoral Research: "Analyzing Forest Degradation Across Biophysical Gradients Using Airborne and Spaceborne LiDAR"

CAHSS Dean's Summer Fellowship ($ 6,000) 2024
For Doctoral Research: "Analyzing Forest Degradation Across Biophysical Gradients Using Airborne and Spaceborne LiDAR"

Department of GES Summer Fellowship ($ 4,000) 2023
For Doctoral Research: "Analyzing Forest Degradation Across Biophysical Gradients Using Airborne and Spaceborne LiDAR"

ITC Excellence Scholarship (€ 36,525) 2019 - 2021
For M.Sc. in Geo-information Science and Earth Observation Program

The Holland Scholarship (€ 10,000) 2019 - 2021
For M.Sc. in Geo-information Science and Earth Observation Program

Erasmus Mundus Scholarship (€ 49,000) 2019
Ranked 14th out of the first selected 20 candidates for MEDfOR M.Sc. Accepted the ITC master's program instead of this program.

TECHNICAL SKILLS

Geospatial Software & Image Processing

- **GIS Software/Platforms:** ArcGIS Pro, ArcGIS Online, QGIS
- **Remote Sensing Suites:** ERDAS IMAGINE, ENVI, eCognition for classification, object-based image analysis, and change monitoring
- **3D & Photogrammetry Tools:** Pix4D, Agisoft Metashape for UAV-based canopy modeling and structure-from-motion workflows
- **Database & Portal Management:** Development and administration of geodatabases and geospatial portals for research and decision-support

Remote Sensing & Geospatial Analysis

- **Satellite Image Analysis:** Multispectral and hyperspectral processing for land cover classification, forest degradation mapping, and change detection
- **SAR & LiDAR Applications:** Use of Synthetic Aperture Radar (ALOS-PALSAR, Sentinel-1) and LiDAR (spaceborne, airborne) for forest structure, biomass, and canopy dynamics
- **Spatial Statistics:** Expertise in environmental modeling and geostatistical analysis.
- **Machine & Deep Learning:** Implement supervised/unsupervised models for land cover mapping, anomaly detection, and predictive analytics.
- **Forest Monitoring Systems:** Integration of multi-sensor, multi-resolution data to assess tropical forest disturbance and recovery

Programming & Computational Environments

- **Languages:** Python, R, Visual Basic
- **Cloud & APIs:** Google Earth Engine (JavaScript & Python APIs), AWS
- **Python Ecosystem:** Proficient with various Python libraries for LiDAR pipelines, SAR analysis, spatial ML, and visualization
- **Statistical Environments:** Advanced R workflows for ecological modeling, uncertainty analysis, and visualization
- **High-Performance Data Handling:** Skilled in parallel processing, AWS S3 workflows, and large-scale raster/vector computation.

PEER-REVIEW SERVICE

2025 - Present Nature
2025 - Present Nature Communications Sustainability
2023 - Present International Journal of Applied Earth Observation and Geoinformation

LEADERSHIP SERVICE

President Mar 2014 – Apr 2015
Green Explore Society, Shahjalal University of Science and Technology

Co-founder & Vice-President Jan 2014 – Dec 2015
Golap-Bandhan Student Association, Shahjalal University of Science and Technology

Treasurer Mar 2013 – Mar 2014
Green Explore Society, Shahjalal University of Science and Technology

General Member Mar 2011 – Dec 2015
KIN, A Voluntary Organization of SUST, Shahjalal University of Science and Technology

General Member Jan 2011 – Dec 2018
Bangladesh Professional Forestry Students Association (BPFSA),
Shahjalal University of Science and Technology